



SparXiv: A Spark-Based Recommender System for arXiv

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Problem Statement:

Research Discovery Needs a Fix

1

Millions of new papers each year

2

Existing tools can't handle large-scale metadata

3

Keyword search misses real relevance

4

Minimal trend analytics

5

Slow or infrequent updates



Dataset Overview: The Foundation

1

Official arXiv metadata

This includes essential information about each paper.

2

3M+ papers (4–6 GB JSONL)

A massive collection of scientific literature.

3

Titles, abstracts, authors, categories, versions

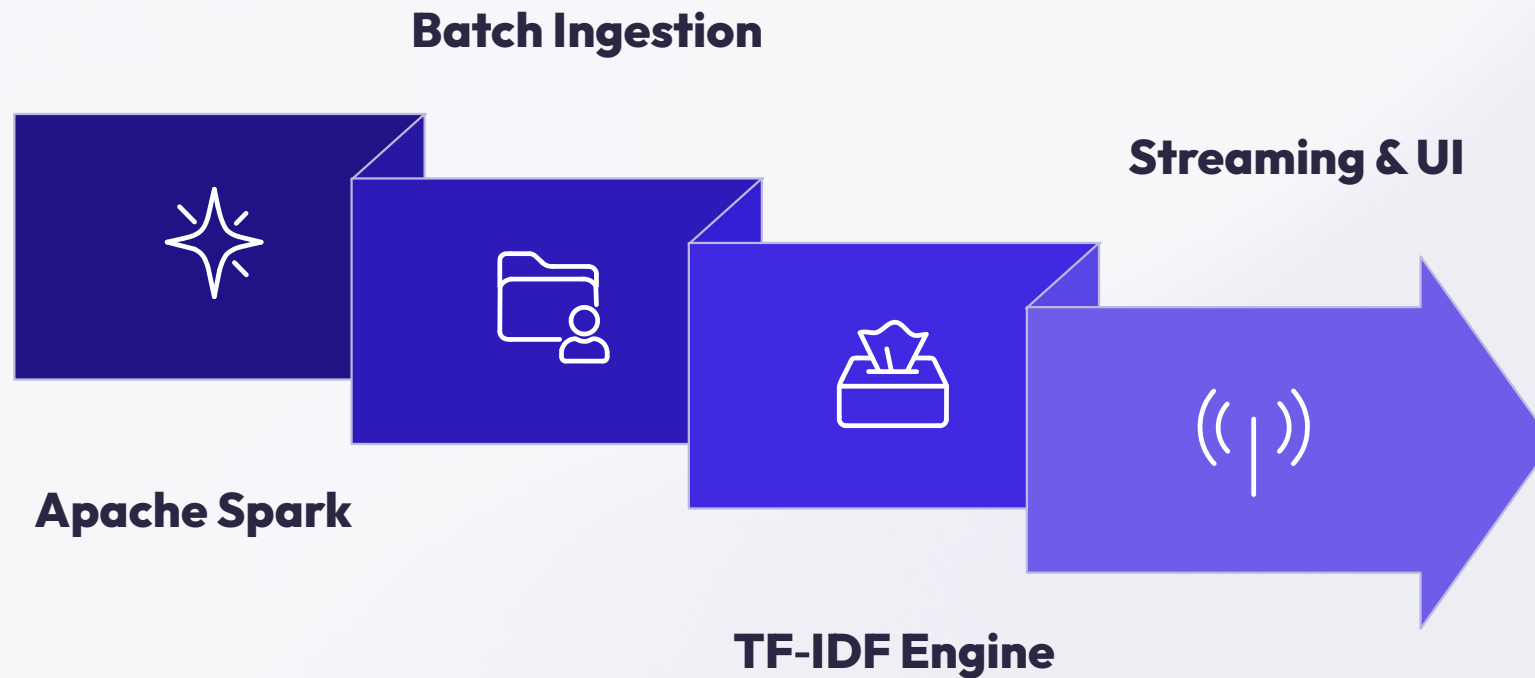
Detailed fields for comprehensive analysis.

4

Weekly updates

Ensuring the dataset remains current and relevant.

System Architecture: Built for Scale and Speed



Batch Data Engine: Reliable, Clean Metadata

01

JSONL to normalized Parquet

Efficient conversion for structured data storage.

02

Consistent, standardized fields

Ensuring uniformity across all metadata.

03

Scalable text cleaning

Handling large volumes of text data effectively.

04

Optimized for millions of records

Designed for high-performance processing.

AI-Powered Similarity Search: Instant, Relevant Recommendations



Distributed TF-IDF vectors



Cosine similarity retrieval



Search by title, abstract, or free text



Analytical Insights: Actionable Research Intelligence



Category co-occurrence maps



Topic growth trends

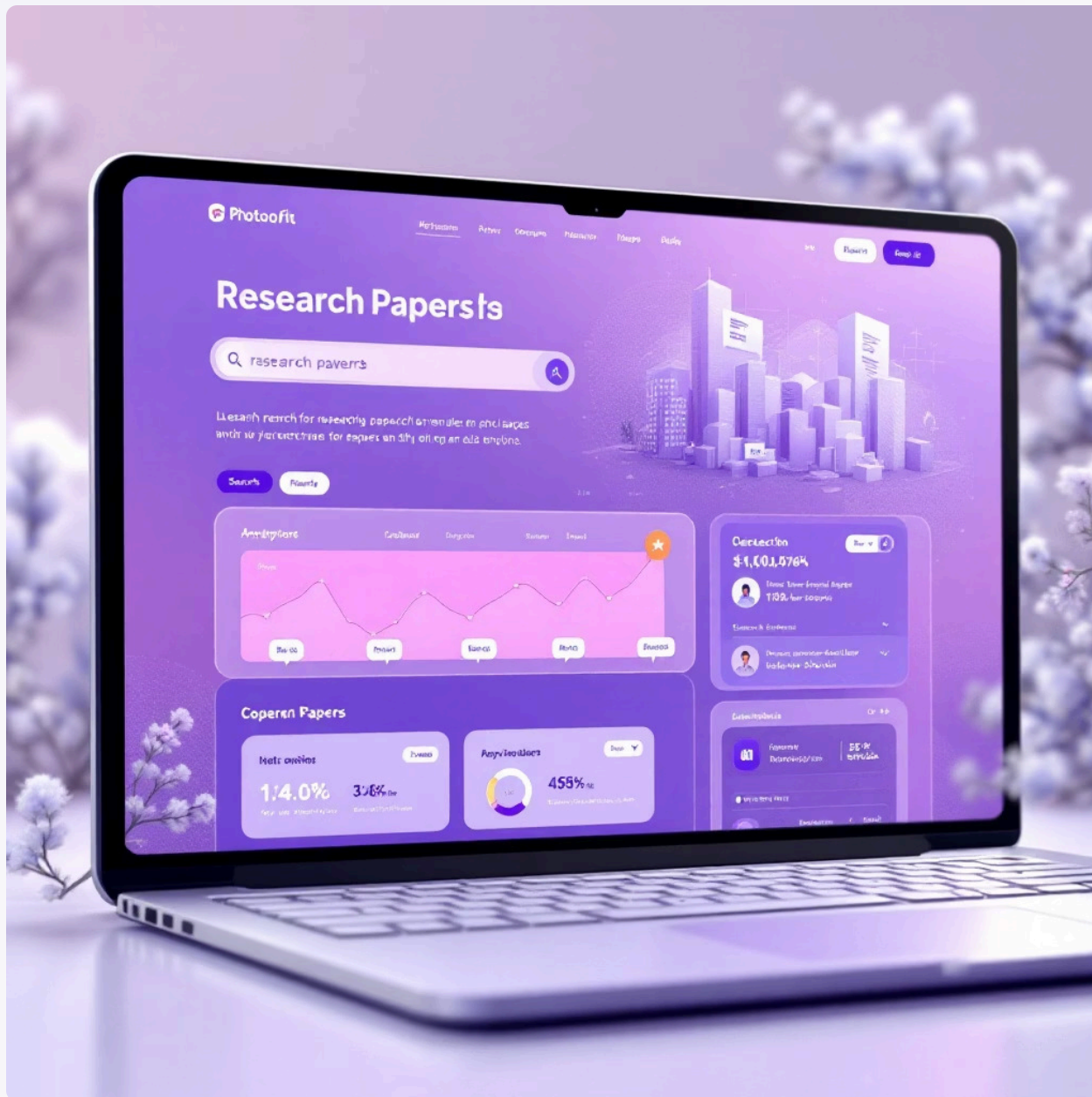


Collaboration and productivity signals

Streaming Analytics: Always Current

- Ingests each new metadata release
- Automatic insights: submissions, DOIs, top categories

Web Application: Ready for Real Users



Interactive similarity search



Built-in analytics dashboards



Ready for deployment



Key Results and Impact: What SparXiv Delivers

1

End-to-end data processing

2

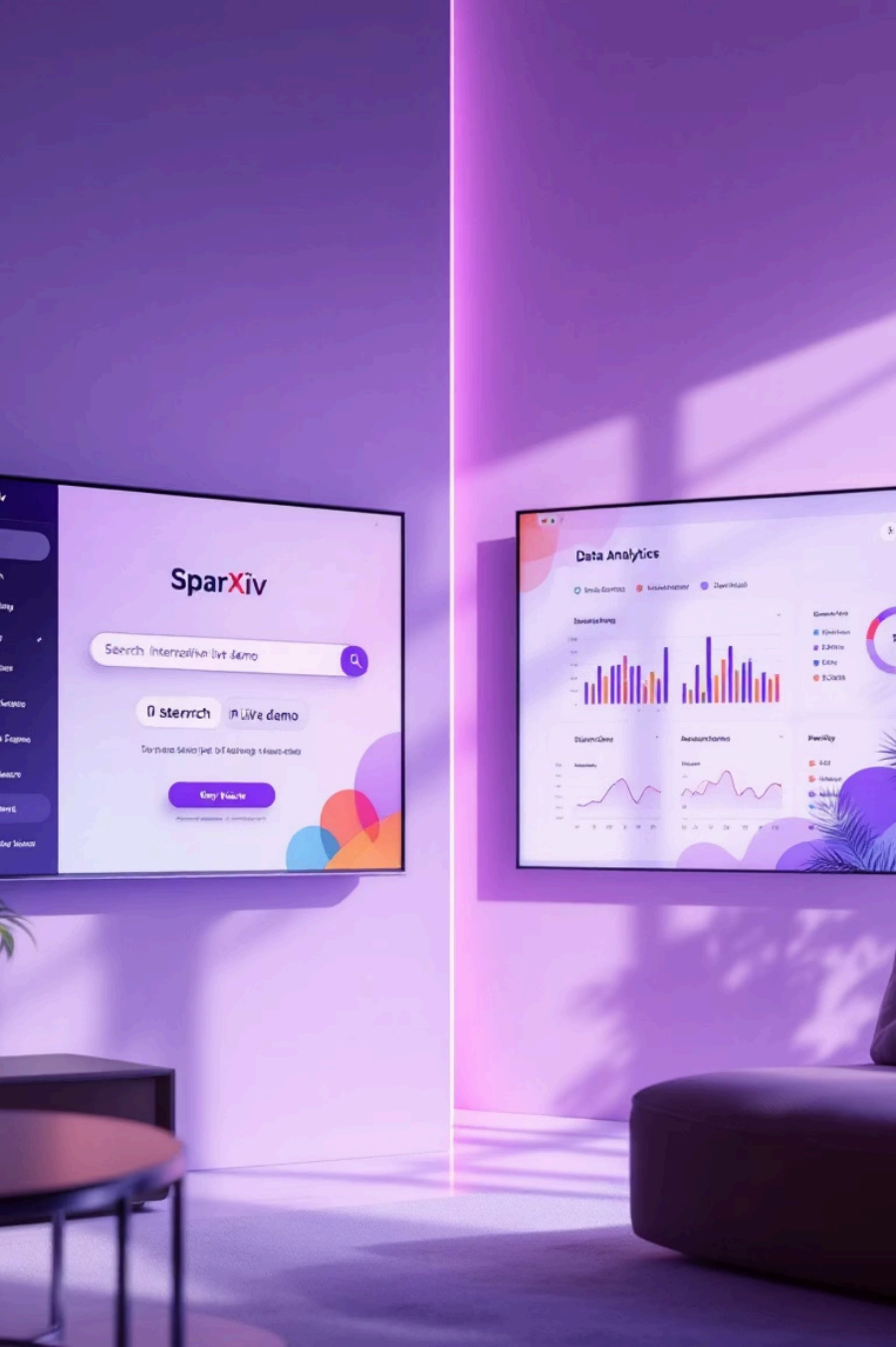
Context-aware recommendations

3

Trend dashboards that matter

4

Weekly dataset refresh



Live Demo

Competitors:

- [arXiv](#)
- [BASE](#)
- [CORE](#)
- [Google Scholar](#)
- [ResearchGate](#)
- [Semantic Scholar](#)
- [ChatGPT Deep Research](#)

Q&A

Any Questions?